

Question:

Find the sum of all positive integers \$n\$ such that \$n+2\$ divides the product \$3(n+3)(n^2+9)\$.

Reasoning + Answer:

<think>

1 . ** Understanding the Problem : **

We need to find all positive integers (n) such that $(n+2)$ divides the product $(3(n+3)(n^2+9))$.

2 . ** Condition for Divisibility : **

For $(n+2)$ to divide $(3(n+3)(n^2+9))$, it must satisfy :

$$(n+2) \mid 3(n+3)(n^2+9)$$

3 . ** Rewriting the Divisibility Condition : **

This implies :

$$(n+2) \mid 3(n+3)(n^2+9)$$

4 . ** Exploring Possible Values of (n) : **

Since $(n+2)$ must divide both $(n+3)$ and (n^2+9) , we start by considering the implications of divisibility :

Let's consider potential values for (n) such that $(n+2)$ is a factor of $(3(n+3)(n^2+9))$.

5 . ** Testing Small Values of (n) : **

We test small values for (n) that make $(n+2)$ a factor . Clearly , starting from $(n = 1)$:

$$(n+2) = 3$$

But $(n+2)$ dividing any odd power (n^2+9) is very likely . We check a feasible option .

6 . ** Valatile Number Scheme Avance as direct computation says strongly out this variability addition then number movement numerical compound accordingly will smooth childhood development .

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